Practical 7: Intraplate volcanism; mantle plumes

On successful completion of this practical, you should be able to:

- Describe the fractionation histories of tholeiitic and alkali basalt suites
- Calculate melt temperatures, primary melt compositions and partial melt fractions using the equations provided (you do not need to memorize the equations)
- Relate differences in mantle temperature and melt fraction to the underlying mantle plume

You will also practice the following skills:

- Solve igneous petrological problems using numerical and graphical methods

Introduction

The composition of magma that is erupted at the Earth's surface depends on:

- The composition of the source mantle
- The amount of partial melting
- The process by which the melting occurred
- The fractional crystallization history of the magma between source and surface.

In this practical we will be investigating the origins of two suites of basaltic rocks from Hawaii.

Part 1 (20 minutes): Fractionation history of alkali and tholeiitic basalts

Table 1 shows representative basalts from two magma suites: one from Pu'u 'Ō'ō and one from Mauna Kea. Pu'u 'Ō'ō is a volcanic cone located on the eastern rift zone of Kilauea volcano. It erupted continuously from January 1983 until April 2018. Mauna Kea is the highest volcano in Hawaii, but it has moved into the post-shield stage of its lifecycle and is currently dormant.

1. One of the magma suites is tholeiitic and the other is alkaline. Which one is which?

   Pu'u O'o is tholeiitic. Mauna Kea is alkaline. Use the TAS plot to check the compositions.

2. Plot variation diagrams of FeO, TiO_2 and K_2O (on the y axis) versus MgO for the samples in the two suites using the templates provided. A plot of CaO vs. MgO is provided for you.

   See plots.

3. Comment on the patterns you observe in the oxide-oxide plots and in the observed phenocryst assemblages (listed in Table 1). You may find the major element compositions of forsteritic olivine, bytownite plagioclase and augite (Table 2) helpful.

   a) Could crystallization of the observed minerals be controlling the evolution of the magma compositions?
   Yes. The compositions of the three most primitive samples in each suite are consistent with olivine fractionation. The more evolved three samples have compositions that are consistent with fractionation of cpx + plagioclase. This is also consistent with the phenocryst assemblages observed in the samples (given in Table 1).

b) Where in the magmatic system do you think the phenocrysts formed?
Most likely they were formed free-floating in a magma chamber, where the cooling rate was slow. To verify this we would need to look at the samples – if this is correct we should observe large euhedral phenocrysts or glomerocrysts.

Part 2 (15 minutes): Liquid thermometry

There are various methods for estimating the temperature of a basalt magma from it chemistry or from the chemistry of its phenocrysts. One of the simplest is the glass (liquid) thermometer of Helz & Thornber (1987), which is dependent only upon the MgO content of the liquid:

$$T(°C) = 26.3MgO^{liq} + 994.4$$

4. Use this equation to calculate magma temperatures for the two Hawaiian magma suites. Add your answers to Table 1.
See Table 1.

5. Comment on the temperature ranges for the two magma suites.

a) How does magma temperature change as fractional crystallization proceeds?
The most Mg-rich samples are the hottest. As crystallization proceeds, temperature decreases. This is what we would expect for fractional crystallization. For a given MgO content, there isn't much difference between the two suites, but this is what we would expect given that we are using a fairly simple liquid thermometer based only on melt MgO content.

b) Do your answers provide any evidence for elevated mantle potential temperature beneath Hawaii?
The most primitive sample in each suite has a temperature of just under 1280 °C. This is very close to the potential temperature of ambient (normal) mantle, which could suggest that the magmas are not unusually hot. However, erupted magmas are always cooler than the mantle potential temperature, because of the latent heat of melting. So, unless the latent heat of melting is extremely small, these samples do suggest that the mantle potential temperature could be hotter than ambient.

Part 3 (30 minutes): Primary magma compositions

Even the most primitive samples in the two magma suites contain olivine phenocrysts. Therefore, these magmas must have undergone fractional crystallization before they erupted – so they are not primary magmas. However, we can use the compositions of the erupted magmas to attempt to constrain the composition(s) of the primary magmas (i.e., the magma composition before any crystallization had occurred).

First, we will use the partitioning of Fe and Mg between melt and olivine to calculate the MgO content of the primary magma.

Under equilibrium conditions, FeO and MgO can be thought of as partitioning into melt and olivine according to the following reaction:

$$MgO_{olivine} + FeO_{melt} = MgO_{melt} + FeO_{olivine}$$

The equilibrium coefficient for this reaction, K_D, is defined as:

$$K_D^{Fe-Mg} = \frac{FeO_{olivine} MgO_{melt}}{MgO_{olivine} FeO_{melt}}$$

For basaltic magma, K_D~0.3.

The concentrations of compatible elements (like Mg and Fe) in the solid phases vary only slowly with the degree of melting, so most mantle minerals have relatively constant Fe/Mg ratios. Mantle olivines have typical compositions around Fo_{90} (that is, 90 mol.% forsterite, 10 mol.% fayalite). This is equivalent to a FeO/MgO mass ratio of about 0.2. We can substitute this value into the equation above in order to calculate the MgO/FeO ratio of a primary melt in chemical equilibrium with mantle olivine:

$$K_D = 0.3 = 0.2 \frac{MgO_{melt}}{FeO_{melt}}$$

Which simplifies to:

$$\frac{MgO_{melt}}{FeO_{melt}} = 1.5$$

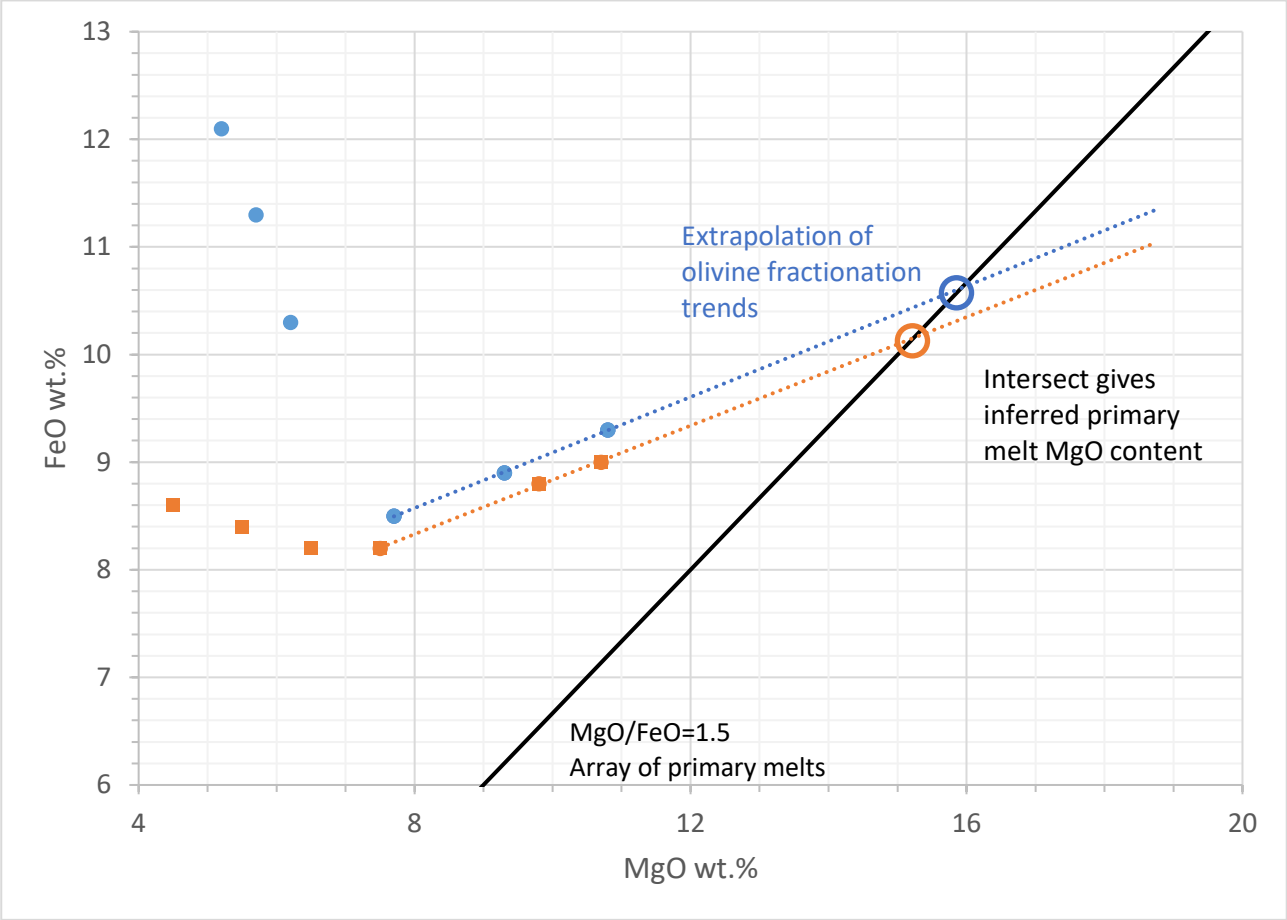
6. Calculate the MgO/FeO ratio of the most primitive basalt compositions from Pu‘u ‘Ō‘ō and Mauna Kea. Are these magmas close to primary compositions?
   MgO/FeO ≈ 1.2 for both primitive basalts. These cannot be not primary magmas.

7. Use the FeO vs. MgO plot of the observed basalt compositions, and the primary melt MgO/FeO ratio (above), to estimate the compositions of the primary magmas. Add your calculated values to Table 3 below.
   Hint: The first step is to plot the compositions of possible primary melts that obey the criterion of being in equilibrium with mantle olivine, using the equation MgO/FeO = 1.5.

   Use FeO and MgO to work out the MgO content of melt in equilibrium with primary melt (i.e. in equilibrium with mantle olivine). Extrapolate the olivine fractionation lines to obtain the inferred primary melt composition. See plot.

Table 3: Primary melt compositions and temperature of Hawaiian basalts.

Primary melt	Pu‘u ‘Ō‘ō	Mauna Kea
MgO wt.%	16.2	15.3
TiO_2 wt.%	2.1	2.5
K_2O wt.%	0.4	0.7
T, °C	1420	1397



8. Now determine the TiO_2 and K_2O contents of the primary melts for each magma suite. Add this information to Table 3.
   Extrapolate the olivine fractionation trends on the TiO_2 vs MgO and K_2O vs MgO plots. Read off the TiO_2 and K_2O contents at the correct MgO for each primary melt composition.

9. Finally, calculate the temperature of the primary melts for each magma suite. Add this information to Table 3.

10. Now interpret your results:

    a) Do your calculated primary melts provide any evidence for elevated mantle potential temperature at Hawaii?
    Yes. The primary melts of both magma suites are at least 100 °C than ambient mantle potential temperature.

    b) Are the primary melt temperatures equivalent to mantle potential temperatures?
    No. The true mantle potential temperature at Hawaii must be even hotter than the calculated primary melt temperatures, because we have not accounted for the latent heat of melting.

Part 4 (20 minutes): The fraction of partial melting

Trace elements behave in predictable ways during partial melting. If we know the partition coefficient *D*, then the fraction of melting *F* can be calculated from the following equation:

$$\frac{C_{melt}}{C^0_{solid}} = \frac{1}{D(1-F)+F}$$

Where C_{melt} is the concentration of the trace element in the melt, and C^0_{solid} is the concentration of the trace element in the original solid (in our case, the mantle peridotite).

We are going to assume that Ti and K behave as perfectly incompatible trace elements during mantle melting.

10. What is the partition coefficient *D* if the trace element is perfectly incompatible?
    If the element is perfectly incompatible then the partition coefficient tends towards zero.

11. In light of your answer, simplify the equation above.
    If D=0 then the *D(1-F)* term vanishes, leaving:

$$\frac{C_{melt}}{C^0_{solid}} = \frac{1}{F}$$

We are going to assume that both magma suites formed from the same type and composition of mantle peridotite. The composition of a mantle peridotite called KLB-1 is provided in Table 1. Using your new equation, and the composition of KLB-1, calculate the fraction of melting required to produce the two primary magmas.

Table 4: Fraction of melting required to produce primary melts beneath Hawaii

	Pu‘u ‘Ō‘ō	Mauna Kea
Ti	F=0.078 (7.8%)	F=0.062 (6.2%)
K	F=0.053 (5.3%)	F=0.029 (2.9%)

12. Now interpret your results:

    a) Why do the Ti and K calculations not yield the same value of F? What assumptions have we made might explain this result?
    We assumed that Ti and K are both perfectly incompatible (i.e., they always partition in the melt) which is a good approximation but not strictly true – small amounts of Ti and K must remain in the solid mantle residue (Ti is more compatible than K in pyroxenes). We also assumed that both the Pu’u O’o and Mauna Kea magmas are produced by melting mantle with the same composition as KLG-1. Perhaps a more K-rich mantle composition would be more appropriate.

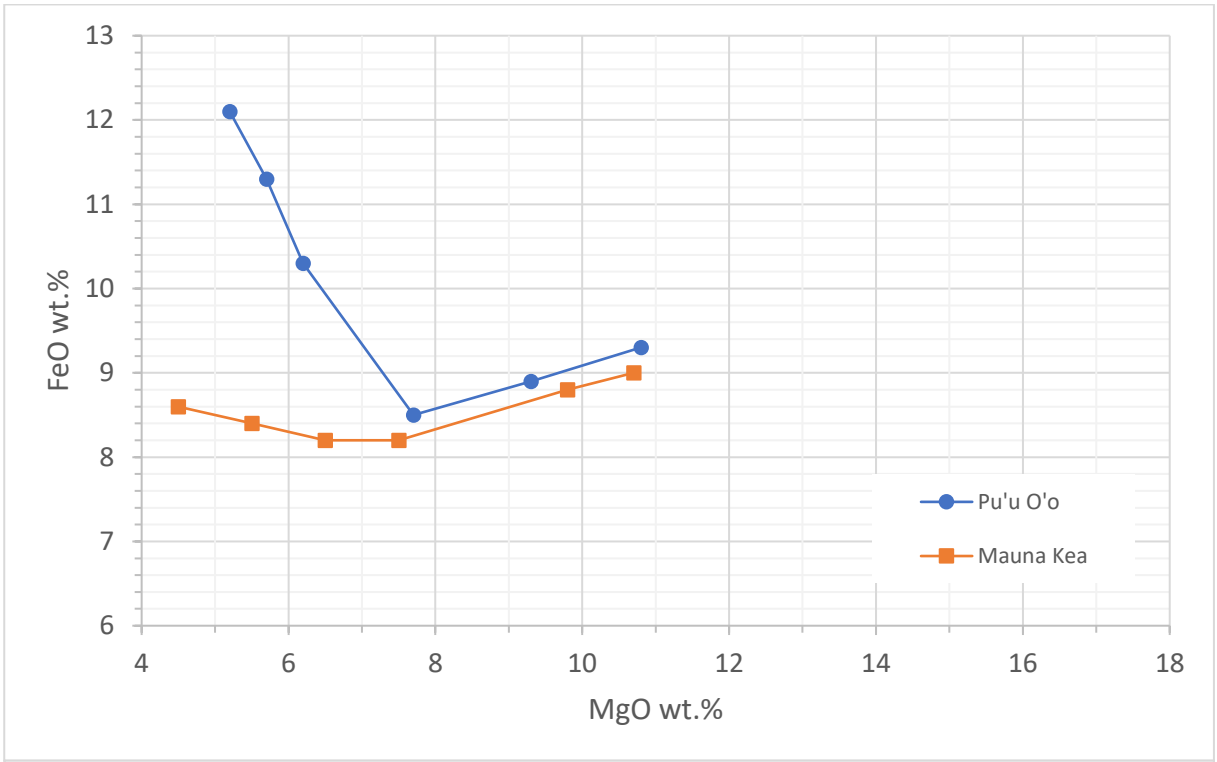
    b) In light of these calculations, sketch the probable locations of Pu’u O’o and Mauna Kea with respect to the Hawaii mantle plume.
    Pu’u O’o must be closest to the plume stem, because the melt fraction is higher and temperature is higher. Mauna Kea is a post-shield volcano, so it is now located away from the plume stem.

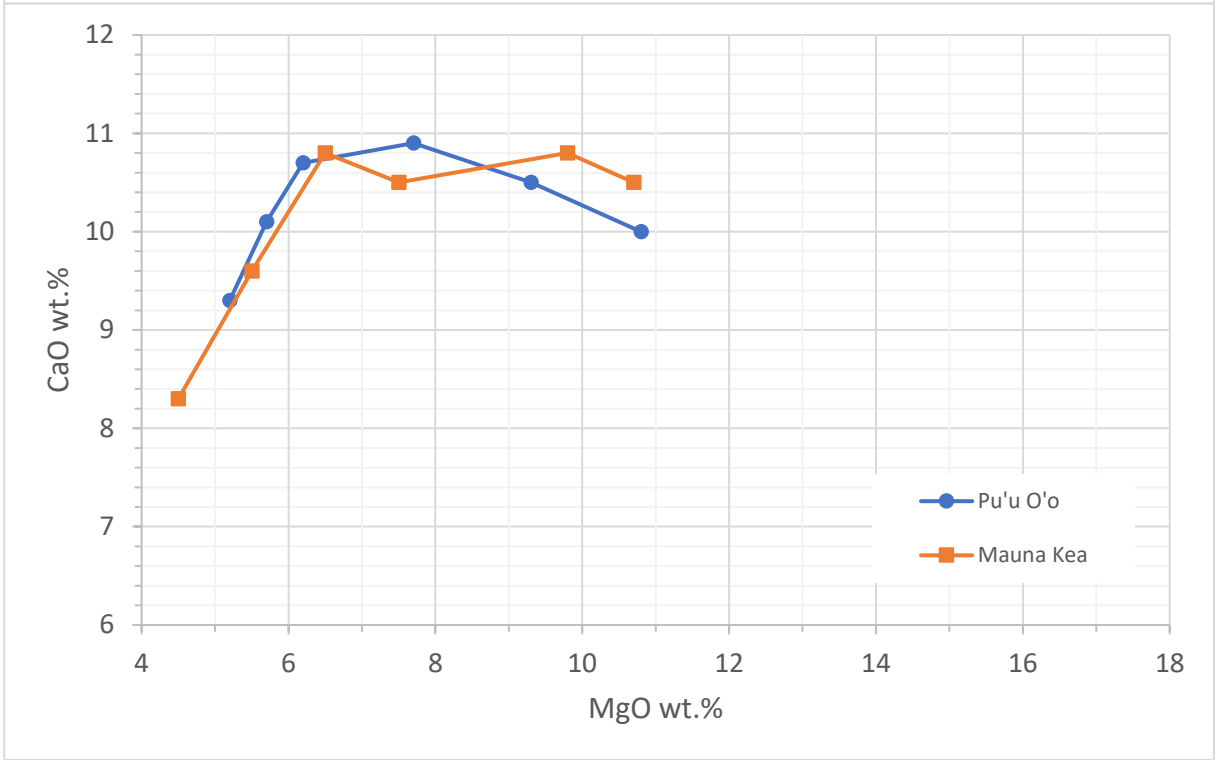
Table 1: Whole-rock compositions and observed phenocryst assemblages for basalt samples from Pu'u O'o and Mauna Kea volcanoes, Big Island, Hawaii.

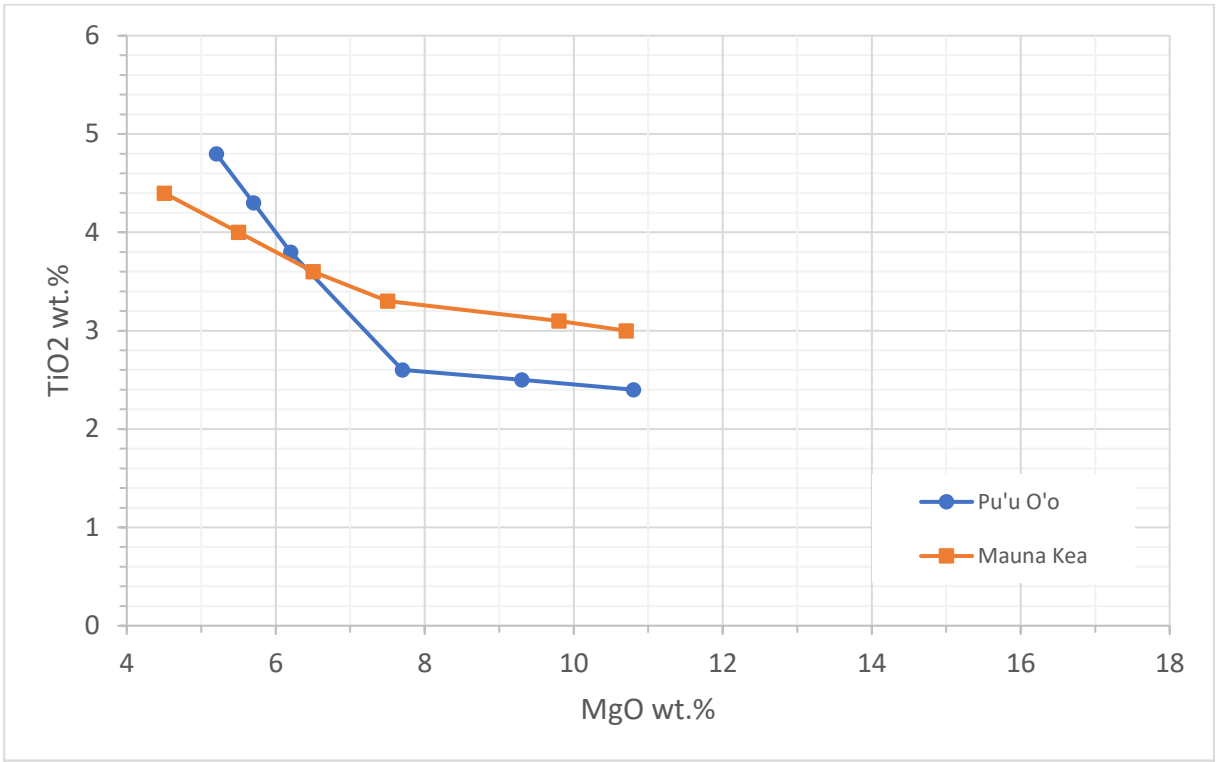
Sample	**Phenocrysts**	**SiO_2**	**TiO_2**	**Al_2O_3**	**FeO**	**MgO**	**CaO**	**Na_2O**	**K_2O**	**T, °C**
Pu'u O'o 1	olivine	50.1	2.4	13.3	9.3	10.8	10.0	2.16	0.43	1278
Pu'u O'o 2	olivine	50.5	2.5	14.0	8.9	9.3	10.5	2.25	0.44	1239
Pu'u O'o 3	olivine	51.1	2.6	14.7	8.5	7.7	10.9	2.36	0.46	1197
Pu'u O'o 4	cpx + plag	51.4	3.8	14.6	10.3	6.2	10.7	2.76	0.66	1157
Pu'u O'o 5	cpx + plag	51.6	4.3	14.4	11.3	5.7	10.1	2.76	0.78	1144
Pu'u O'o 6	cpx + plag	51.7	4.8	14.3	12.1	5.2	9.3	3.34	0.86	1131
Mauna Kea 1	olivine	44.9	3.0	14.1	9	10.7	10.5	3.12	0.79	1276
Mauna Kea 2	olivine	45.1	3.1	14.5	8.8	9.8	10.8	3.20	0.81	1252
Mauna Kea 3	olivine	45.4	3.3	15.6	8.2	7.5	10.5	4.55	0.86	1192
Mauna Kea 4	cpx + plag	45.2	3.6	16.4	8.2	6.5	10.8	5.15	0.94	1165
Mauna Kea 5	cpx + plag	44.2	4.0	17.7	8.4	5.5	9.6	5.78	1.08	1139
Mauna Kea 6	cpx + plag	41.0	4.4	18.8	8.6	4.5	8.3	6.35	1.21	1113
KLB-1 peridotite		44.5	0.16	2.6	8.1	39.2	3.4	0.30	0.02	

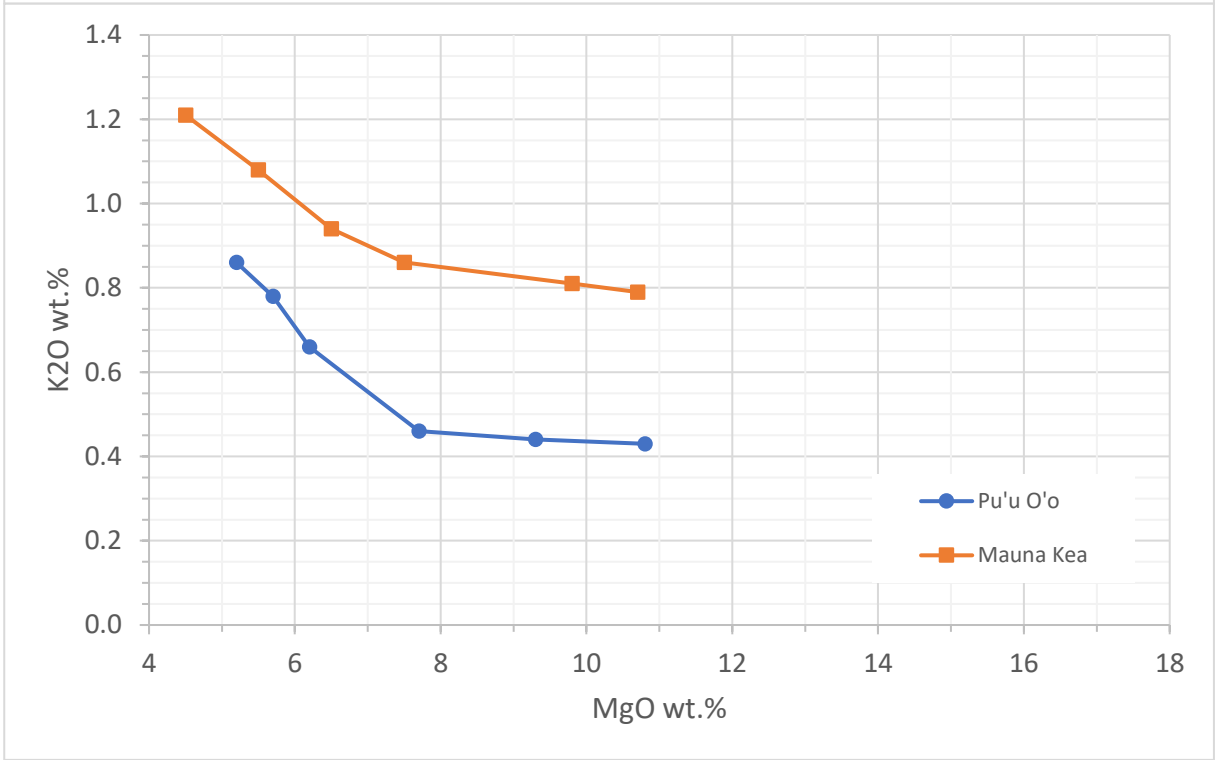
Table 2: Representative compositions of olivine, plagioclase and clinopyroxene in Hawaiian basalts.

	Olivine, Fo_{85}	Plagioclase, An_{76}	Augite
MgO	46.3	-	17.6
FeO	13.9	-	7.1
CaO	0.2	15.4	19.0
TiO_2	-	-	0.5
K_2O	-	0.2	-


FeO wt.%
MgO wt.%
Pu'u O'o
Mauna Kea


CaO wt.%
MgO wt.%
Pu'u O'o
Mauna Kea


TiO2 wt.%
MgO wt.%
Pu'u O'o
Mauna Kea


K2O wt.%
MgO wt.%
Pu'u O'o
Mauna Kea